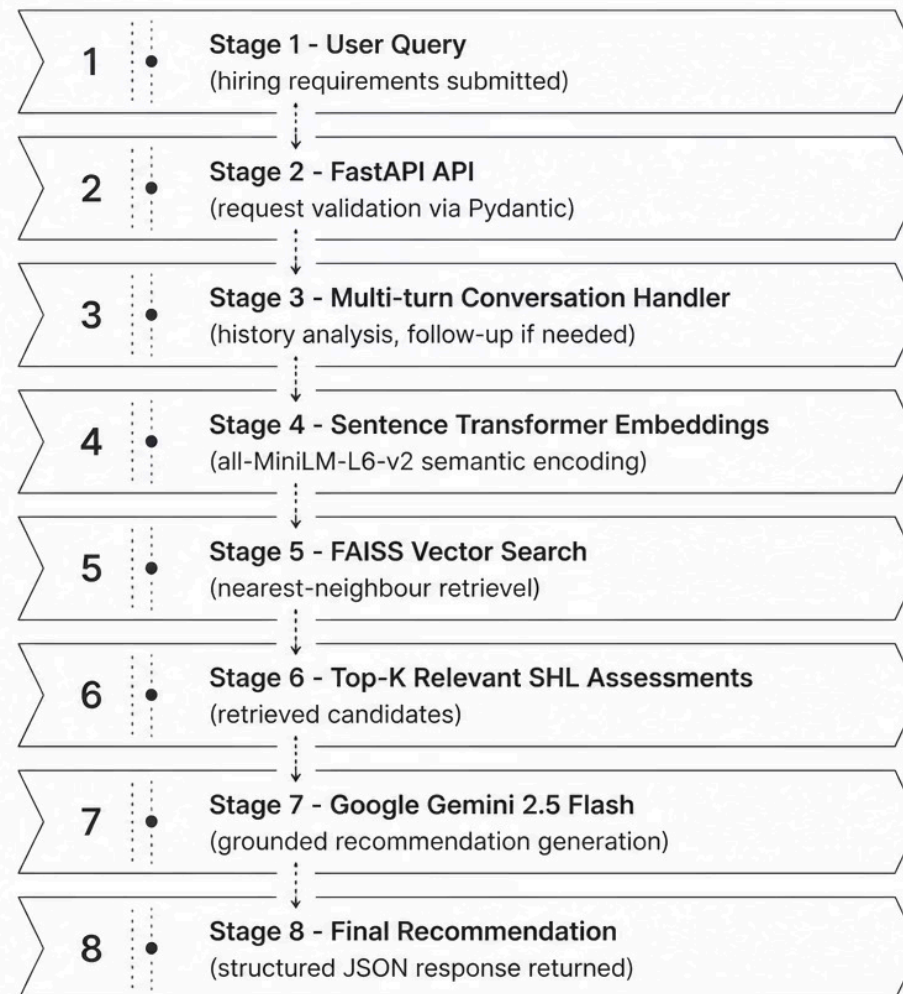


SHL AI Assessment Recommendation Agent

Prepared for: SHL Generative AI Internship Assignment · Developer: Divyash Saxena

An AI-powered recommendation system that suggests suitable SHL assessments based on hiring requirements. Built on Retrieval-Augmented Generation (RAG) and exposed via FastAPI REST API.

System Architecture



Stateless API design — conversation history is passed in every request. Follow-up questions are triggered when hiring context is incomplete.

Technology Stack & Engineering Decisions

Component	Technology	Rationale
API	FastAPI	High-performance async REST
Embedding Model	Sentence Transformers (all-MiniLM-L6-v2)	Semantic similarity
Vector DB	FAISS	Fast nearest-neighbour retrieval
LLM	Google Gemini 2.5 Flash	Grounded recommendations
Validation	Pydantic	Schema validation
Configuration	python-dotenv	Secure key management

Key Decisions & Challenges

- Modular architecture with clean separation of concerns
- Config-driven development, reusable retrieval module
- REST API design with easy deployment
- Challenges: malformed JSON, semantic embeddings, FAISS indexing, prompt engineering, API compatibility, multi-turn design

Future Enhancements

- Hybrid retrieval (BM25 + FAISS)
- Docker containerization
- Persistent conversation memory
- Assessment re-ranking
- Streaming responses
- Confidence scoring



The system combines semantic retrieval with Google Gemini 2.5 Flash to deliver grounded, accurate SHL assessment recommendations – satisfying all assignment requirements.